# Renal Reference — Low FENa Exceptions

## The “exceptions” they test (low FeNa doesn’t always mean prerenal)

Even though **FeNa <1% usually suggests prerenal**, it can also be **<1% in intrinsic AKI**, especially:

1. **Early sepsis**
2. **Contrast-associated / contrast-induced AKI**
3. **Rhabdomyolysis**
4. **Glomerulonephritis (early)**

### Practical board takeaway

In real life and on boards, often more important than FeNa:

- **history + exam**
- **urine sediment**
- **response to fluids**

## 1. Exceptions header

**WHAT YOU SEE**

Title 'The exceptions they test (low FENa doesn't always mean prerenal)'

**WHAT IT ENCODES**

Core trap: FENa <1% does NOT prove prerenal AKI. FENa = (UNa × PCr)/(PNa × UCr) × 100. The rule of thumb is FENa <1% = prerenal, FENa >2% = ATN, but several intrinsic processes give a falsely low FENa because the tubules are still able to (or are being driven to) avidly reabsorb sodium. FEUrea <35% is the better discriminator when the patient is on diuretics, since diuretics force natriuresis and falsely raise FENa.

**BOARD TRAP**

WRONG: using FENa as a yes/no test for prerenal vs intrinsic AKI. FENa is only interpretable in oliguric AKI off diuretics, and it has documented exceptions in both directions. Discriminator: integrate FENa with sediment, history, and FEUrea — never let a single number override the clinical picture.

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## 2. Low FENa usually prerenal

**WHAT YOU SEE**

Line: 'FENa <1% usually suggests prerenal, but can be <1% in intrinsic AKI'

**WHAT IT ENCODES**

Baseline rule then the exception. In true prerenal azotemia the kidney senses low effective volume → RAAS + ADH drive maximal Na and water reabsorption → UNa <20, FENa <1%, urine osm >500, BUN:Cr >20:1, bland sediment. The exceptions below show intrinsic injuries that also produce FENa <1% early or with intact tubular function, so a low number alone cannot exclude them.

**BOARD TRAP**

WRONG: a BUN:Cr >20 plus FENa <1% means 'just give fluids and the kidney will recover.' Hepatorenal syndrome, early GN, contrast and pigment nephropathy all mimic the prerenal profile yet may not respond to volume. Discriminator: reassess after a fluid challenge — failure to improve flags an intrinsic or hemodynamic process.

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## 3. 1. Early sepsis

**WHAT YOU SEE**

Numbered item 1 'Early sepsis'

**WHAT IT ENCODES**

In early sepsis-associated AKI, intense neurohormonal activation and renal vasoconstriction keep the tubules conserving sodium → FENa <1% even though microvascular/inflammatory injury (the actual mechanism of septic AKI) is already underway. As septic ATN becomes established the FENa typically climbs above 2%.

**BOARD TRAP**

WRONG: a septic patient with FENa <1% just needs volume and has no intrinsic injury. Early septic AKI is intrinsic despite the low FENa, and over-resuscitation can cause venous congestion that worsens kidney function. Discriminator: in distributive shock, pursue source control + balanced resuscitation; don't be reassured by a low FENa.

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## 4. 2. Contrast-associated AKI

**WHAT YOU SEE**

Item 2 'Contrast-associated / contrast-induced AKI'

**WHAT IT ENCODES**

Iodinated contrast causes intense medullary vasoconstriction (via endothelin/adenosine, reduced NO) plus direct tubular toxicity. The vasoconstrictive, low-flow physiology makes the kidney avidly reabsorb sodium → FENa often <1% despite this being an intrinsic/tubular insult. Creatinine rises 24–48h, peaks 3–5 days, recovers by 7–10 days. Prevention is isotonic saline; N-acetylcysteine and bicarbonate are NOT proven (PRESERVE trial negative).

**BOARD TRAP**

WRONG: a low FENa 2 days after a CT angiogram means the AKI is prerenal from contrast-related NPO status. Contrast nephropathy is an intrinsic insult that classically produces a low FENa, so the number misleads. Discriminator: timing (rise within 24–48h of contrast) + the procedure history clinch contrast-associated AKI.

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## 5. 3. Rhabdomyolysis

**WHAT YOU SEE**

Item 3 'Rhabdomyolysis'

**WHAT IT ENCODES**

Pigment (myoglobin) nephropathy injures tubules via three hits: renal vasoconstriction, direct heme toxicity, and intratubular cast obstruction. The early vasoconstrictive phase drives avid Na reabsorption → FENa is frequently <1% despite established tubular injury. Clues: CK often >5,000 (frequently >15,000–20,000), heme-positive dipstick with FEW/no RBCs on micro, hyperkalemia, hyperphosphatemia, hypocalcemia, high uric acid. Treat with aggressive IV isotonic fluids targeting urine output ~200–300 mL/h.

**BOARD TRAP**

WRONG: pigment/rhabdo AKI 'always has a high FENa' so a low FENa rules it out. Rhabdomyolysis is a classic cause of a deceptively low FENa because of early renal vasoconstriction. Discriminator: heme-positive dipstick with minimal RBCs on microscopy + a markedly elevated CK is pigment nephropathy regardless of the FENa value.

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## 6. 4. Glomerulonephritis (early)

**WHAT YOU SEE**

Item 4 'Glomerulonephritis (early)'

**WHAT IT ENCODES**

In early acute glomerulonephritis the lesion is glomerular while the tubules remain intact and avidly reabsorb sodium (often with salt-and-water retention causing HTN/edema) → FENa <1% with active glomerular disease. The sediment, not the FENa, gives it away: dysmorphic RBCs, RBC casts, and sub-nephrotic-to-nephrotic proteinuria.

**BOARD TRAP**

WRONG: FENa <1% with hematuria points to a stone or prerenal state, not the glomerulus. Early GN classically gives a low FENa because the tubules are healthy. Discriminator: RBC casts and dysmorphic RBCs localize to the glomerulus and mandate GN serologies/biopsy, overriding the low FENa.

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## 7. Practical board takeaway

**WHAT YOU SEE**

Header 'Practical board takeaway'

**WHAT IT ENCODES**

When FENa is misleading (sepsis, contrast, rhabdo, early GN — and also hepatorenal syndrome, where splanchnic vasodilation triggers extreme renal Na avidity with FENa often <0.1%), weight the rest of the data: history/exam, urine sediment, and the response to a volume challenge. FENa is also invalid in CKD, with diuretics, or in non-oliguric states.

**BOARD TRAP**

WRONG: trusting FENa over the whole clinical gestalt. In hepatorenal syndrome FENa can be near zero yet fluids alone won't fix it (treat with albumin + vasoconstrictors like terlipressin/norepinephrine or midodrine-octreotide). Discriminator: a near-zero FENa in advanced cirrhosis without another cause = HRS, not simple prerenal.

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## 8. History + exam

**WHAT YOU SEE**

Bullet 'history + exam'

**WHAT IT ENCODES**

Clinical context outweighs FENa for localizing AKI: recent contrast, nephrotoxic drugs, crush/immobilization/statin (rhabdo), volume status (orthostatics, JVP, skin turgor), cirrhosis stigmata, sepsis source, and the medication list. These reframe what a given FENa means.

**BOARD TRAP**

WRONG: skipping the med and exposure history because the labs 'fit prerenal.' A normal-appearing FENa can hide aminoglycoside, contrast, or NSAID injury that the history reveals. Discriminator: the timeline of exposures usually localizes AKI better than any single index.

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## 9. Urine sediment

**WHAT YOU SEE**

Bullet 'urine sediment'

**WHAT IT ENCODES**

Sediment localizes injury better than FENa in the exception cases: bland/hyaline = prerenal, muddy-brown granular casts/RTE cells = ATN, RBC casts/dysmorphic RBCs = GN, WBC casts/sterile pyuria = AIN, heme+ without RBCs = pigment. A microscopy read can override a misleading FENa.

**BOARD TRAP**

WRONG: ordering FENa but not looking at the sediment yourself. Automated UA can miss casts; an examiner expects you to correlate the spun sediment with the index. Discriminator: muddy-brown casts with FENa <1% still means ATN — believe the casts.

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## 10. Response to fluids

**WHAT YOU SEE**

Bullet 'response to fluids'

**WHAT IT ENCODES**

A timed response to a fluid challenge is the functional test prerenal azotemia should improve (creatinine falls, urine output rises) within 24–72h of restoring volume, whereas ATN, contrast, pigment, GN, and HRS do not correct with fluids alone.

**BOARD TRAP**

WRONG: continuing to bolus fluids when creatinine isn't budging because the FENa said prerenal. Non-response to adequate volume reclassifies the AKI as intrinsic/hemodynamic and should stop the fluids (avoid volume overload). Discriminator: failure to improve after a fair fluid trial is itself diagnostic information.

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